

Lecture Summary: Multi-Agent Reinforcement Learning (MARL)

1. Motivation

In classical RL, the environment is stationary and external. In *multi-agent* settings, other agents *are* part of the environment: they learn, adapt, and respond. Thus, single-agent RL assumptions break down, and game-theoretic ideas become essential.

2. Game Theory Basics

One-Shot Games

A normal-form game consists of:

- players $1, \dots, n$,
- action sets A_i ,
- utility functions $u_i : A \rightarrow \mathbb{R}$ where $A = A_1 \times \dots \times A_n$.

Best response. Given actions of others a_{-i} ,

$$BR_i(a_{-i}) = \arg \max_{a_i \in A_i} u_i(a_i, a_{-i}).$$

Dominant strategy. Action a_i *weakly dominates* a'_i if

$$u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i}) \quad \forall a_{-i}.$$

If such an action exists, it is optimal regardless of the opponent's behavior.

Solution Concepts

Dominant Strategy Equilibrium (DSE). Each player's action is a dominant strategy.

Nash Equilibrium (NE). An action profile $a = (a_1, \dots, a_n)$ is a NE if every player is playing a best response:

$$u_i(a_i, a_{-i}) \geq \max_{a'_i} u_i(a'_i, a_{-i}).$$

Mixed-Strategy NE. Each player uses a distribution $p_i \in \Delta(A_i)$ over actions. A profile $p = (p_1, \dots, p_n)$ is NE if:

$$u_i(p_i, p_{-i}) \geq u_i(a_i, p_{-i}) \quad \forall a_i.$$

Nash's Theorem. Every finite game has a mixed-strategy Nash equilibrium.

Correlated and Coarse-Correlated Equilibria

Mixed NE require independent sampling across players. *Correlated* equilibria allow a joint distribution D over actions.

Correlated equilibrium (CE). No player benefits from deviating given the recommended action a_i :

$$\mathbb{E}_{a \sim D}[u_i(a_i, a_{-i})] \geq \mathbb{E}_{a \sim D}[u_i(a'_i, a_{-i}) | a_i].$$

Coarse CE (CCE). Relaxed condition: no player benefits by switching to a fixed alternative action.

Hierarchy.

$$\text{DSE} \subset \text{PSNE} \subset \text{MSNE} \subset \text{CE} \subset \text{CCE}.$$

3. Zero-Sum Games and Minimax

For two-player zero-sum games ($u_1 = -u_2$), the utility can be represented by a matrix U .

Minimax Theorem (Von Neumann)

$$\min_q \max_p p^\top U q = \max_p \min_q p^\top U q.$$

The value of the game is v^* , and any minimax pair (p^*, q^*) is a Nash equilibrium.

This generalizes to continuous-action games when $u(x, y)$ is convex in x and concave in y .

4. Markov (Stochastic) Games

Definition. A two-player zero-sum Markov game consists of:

$$(S, A_1, A_2, P, R),$$

where both players act simultaneously at each step.

Policies. Each player has a Markov policy $\pi_i : S \rightarrow \Delta(A_i)$.

Value of policy pair. For discounted infinite horizon,

$$V^{\pi_1, \pi_2}(s) = \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^{t-1} R(s_t, a_{1,t}, a_{2,t}) \mid s_1 = s \right].$$

Minimax for Markov Games (Shapley)

There exists a value function V^* such that

$$V^*(s) = \max_{\pi_1} \min_{\pi_2} V^{\pi_1, \pi_2}(s) = \min_{\pi_2} \max_{\pi_1} V^{\pi_1, \pi_2}(s).$$

Any minimax pair of policies is a Nash equilibrium.

5. Bellman Equations for Markov Games

Value form.

$$V^*(s) = \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} \mathbb{E}_{a_1 \sim p, a_2 \sim q} [R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V^*].$$

Q-form.

$$\begin{aligned} Q^*(s, a_1, a_2) &= R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V^*(s'), \\ V^*(s) &= \max_p \min_q p^\top Q^*(s) q. \end{aligned}$$

Thus, each state defines a *matrix game* whose Nash value determines $V^*(s)$.

6. Dynamic Programming for Markov Games

Finite-horizon backward induction:

$$V_k^*(s) = \max_p \min_q \mathbb{E}_{a_1, a_2} [R(s, a_1, a_2) + \gamma V_{k-1}^*(s')], \quad V_0^* = 0.$$

7. Nash Q-Learning

Hu & Wellman (2003) generalize Q-learning to two-player zero-sum Markov games.

Algorithm

- Observe s_t , take action pair $a_t = (a_{1,t}, a_{2,t})$.
- Receive reward r_t and next state s_{t+1} .
- Compute the Nash value of the next-state matrix game:

$$V_{t+1} = \max_p \min_q p^\top Q_t(s_{t+1}) q.$$

- Update:

$$Q_{t+1}(s_t, a_t) \leftarrow (1 - \alpha_t) Q_t(s_t, a_t) + \alpha_t (r_t + \gamma V_{t+1}).$$

Convergence

If $\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$, and all state-action pairs are visited infinitely often, then:

$$Q_t \rightarrow Q^*.$$

8. Algorithms for Finding Equilibria

Common iterative schemes:

- **Self-play:** each player best-responds to the other (may cycle).
- **Fictitious play:** each player best-responds to opponent's empirical distribution. Converges asymptotically in zero-sum games.
- **Double oracle:** iteratively expand strategy sets with best responses.
- **Gradient / Mirror descent:** continuous updates for mixed strategies.

Challenges.

- Best responses may *not* be Markovian even when optimal policies are.
- Reduction to normal-form games causes exponential blowup.
- MARL is generally non-stationary and violates single-agent assumptions.

9. Key Takeaways

- Multi-agent RL combines sequential decision-making with game theory.
- Nash equilibria, correlated equilibria, and minimax concepts are foundational.
- Markov games extend MDPs to competitive multi-agent settings with strong guarantees.
- Nash Q-learning generalizes value-based RL to adversarial environments.
- Many MARL algorithms face instability due to non-stationarity and equilibrium computation difficulty.